

Nathan Ngo

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PROFESSIONAL SUMMARY

Software engineer with experience building full-stack web applications using React, Node.js, Python, and Firebase. Developed real-time collaborative systems, REST APIs, and interactive web applications with a focus on scalability, performance, and user experience.

TECHNICAL SKILLS

- **Languages:** C++, C#, Python, Java, JavaScript, SQL, AutoHotkey
- **Frontend:** React, WPF, HTML/CSS, Material UI, Chakra UI
- **Backend:** Node.js, Express.js, REST APIs, Firebase, Firestore

PROJECTS

Windows Process Monitor

2026

- Built a native Windows system monitor with a live hierarchical process tree displaying CPU, memory, disk I/O, thread count, and handle count for hundreds of processes, refreshed every second using MVVM and incremental **ObservableCollection** synchronization.
- Leveraged **P/Invoke** and **WMI** to collect parent-child process relationships, disk activity metrics, command-line arguments, and hardware information not exposed through standard .NET APIs.
- Developed five real-time performance charts (CPU, Memory, Disk, Network, GPU) using custom **WPF** Canvas rendering, auto-scaling axes, and rolling 2-minute historical data retention at 1-second intervals.

Real-Time Collaborative Notes Platform

2025

- Built a full-stack collaborative note-taking platform supporting **real-time multi-user document** editing using React, Express.js, and Firestore.
- Implemented **low-latency synchronization** and **conflict handling** to maintain document consistency during concurrent editing sessions.
- Developed RESTful APIs for note management, user authentication, and collaboration workflows while enforcing role-based access controls.

Music-Based Typing Game

2024

- Built an interactive music-based typing game integrating YouTube and Spotify APIs to deliver dynamic song content and real-time gameplay
- Designed and implemented a **responsive user interface** using React, Chakra UI, and React Router to improve usability and navigation
- Implemented **client-side state management** with Zustand to improve application maintainability

WORK EXPERIENCE

Technical Support Specialist

Sept. 2019 – May 2020

University of Massachusetts - Lowell

Lowell, MA

- Resolved hardware, software, and network issues for students, faculty, and staff in a high-volume support environment
- Delivered technical support to diverse users, translating complex technical concepts into **clear, user-friendly solutions**
- Managed and prioritized support requests through ticketing systems, phone, and in-person interactions

Private Mathematics Tutor

Sept. 2024 - May. 2025

Freelance

Remote

- Provided one-on-one remote tutoring to students in Algebra, Geometry, Pre-Calculus, and Calculus I/II
- Assisted students with homework, test preparation, and coursework, helping improve understanding and academic performance

EDUCATION

University of Massachusetts - Lowell

2019 - 2023

B.S. in Computer Science, Minor in Mathematics

Lowell, MA